HACE: Addressing Viability Failure in Sequential Reinforcement Learning with Homeostatic Auxiliary Rewards

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Abstract

Continual reinforcement learning (CRL) research has focused almost exclusively on *catastrophic forgetting*—the tendency for neural networks to lose old skills when adapting to new tasks. In embodied settings where agents manage shared physiological resources, we identify a complementary failure mode: *viability failure*, wherein agents deplete internal resources and die before they can learn a new task. Established CRL techniques such as Elastic Weight Consolidation (EWC) and Experience Replay (ER) are structurally unable to address viability failure because it is a reward-level survival problem, not a parameter-preservation problem. We propose Hace (Homeostatic Auxiliary for Continual Environments), a task-invariant auxiliary reward based on homeostatic drive reduction that incentivizes energy management without task-specific engineering. On a 10-task sequential gridworld benchmark with 9 agent variants and 2 boundary modes, Hace agents achieve 3× higher boundary solvability and retain 3× more energy at task transitions than all CRL baselines, which perform no better than unprotected baselines on viability metrics. Combining Hace with EWC addresses both viability and forgetting, confirming the two failure modes are orthogonal.

1 Introduction

Reinforcement learning agents deployed in continual settings must adapt to changing objectives without losing previously learned capabilities. The dominant research paradigm in continual RL (CRL) frames this challenge almost exclusively as *catastrophic forgetting*: the tendency for parameters to overwrite old knowledge during new-task training [5, 8, 9]. A rich toolkit has emerged—Elastic Weight Consolidation (EWC [5]), Experience Replay (ER [6]), progressive architectures [7]—all evaluated on whether old-task performance degrades after training on new ones.

We argue that this focus on forgetting obscures an equally critical failure mode in *embodied* sequential settings: **viability failure**. When an environment imposes shared physiological constraints—such as an internal energy variable that depletes with each action—a task transition can be lethal. An agent completing Task T_i with minimal remaining energy arrives at Task T_{i+1} without the resources to survive long enough to learn. No amount of weight regularization or replayed transitions can save an agent that dies in its first three steps.

Consider a concrete example. An agent trained on Conservation (high step cost, must be efficient) transitions to Sprint (very short time limit, must be fast). Under carry-over boundary conditions—where energy persists between tasks—the agent arrives at Sprint with critically low energy and no time to explore. It dies immediately. EWC preserves the Conservation weights perfectly, but those weights are useless if the agent cannot survive.

This paper makes three contributions. First, we formally identify **viability failure** as a distinct failure mode in embodied CRL, separate from catastrophic forgetting, and show that traditional CRL methods cannot address it (Section 3). Second, we propose **Hace**, a task-invariant auxiliary reward based on homeostatic drive reduction that incentivizes energy management across all tasks without task-specific engineering. Third, we design a **10-task sequential gridworld benchmark** with 9 agent variants, 3 task orderings, and 2 boundary modes, enabling systematic study of both failure modes (Section 4). Our experiments (Section 5) show that Hace maintains viability across task transitions while traditional CRL methods fail entirely on survival metrics.

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2 Related Work

Continual RL.. EWC [5] constrains important parameters via a Fisher-weighted penalty; ER [6] interleaves old transitions during new-task training; progressive networks [7] freeze old columns and add capacity per task. These methods have been benchmarked on Continual World [9] and CORA [10], where episode termination is unaffected by resource constraints. Our work highlights that this evaluation paradigm systematically misses viability failure.

Homeostatic RL.. Keramati and Gutkin [1] formalize reward as deviation reduction from an internal setpoint, producing risk-averse and anticipatory behaviors. Yoshida [2, 3] extends this to deep RL and connects it to meta-RL. Konidaris and Barto [4] propose adaptive motivational systems. However, all prior work studies single or slowly shifting environments and does not test whether homeostatic objectives improve sequential task learning.

Transfer and auxiliary rewards. Successor features [11] decompose value into reusable dynamics and task-specific weights. Tomov et al. [12] observe that humans reuse policies across tasks and that task descriptions could be grounded in physiological states—an insight HACE operationalizes. Unlike curiosity [14] or pixel control [13], HACE targets *viability maintenance* rather than exploration or representation learning.

3 Viability Failure and HACE

3.1 Sequential Task Setting

We consider an agent learning a sequence of tasks $T_1, \ldots, T_n$ sharing a common action space, an internal energy variable $E_t \in [0, E_{\max}]$ with per-step cost c_{step}, and a terminal condition $E_t \leq 0 \implies$ death. Tasks differ in grid layout, hazards, food sources, walls, and energy economics (Section 4). At task boundaries, the agent retains its network parameters. Energy is handled by the *boundary mode*: **reset** reinitializes energy to each task's default, while **carryover** preserves the terminal energy from the previous task. Carryover is the ecologically realistic setting where viability failure is most acute.

3.2 Viability Failure

An agent experiences viability failure at boundary $T_i \rightarrow T_{i+1}$ when its expected terminal energy after Task T_i is insufficient to sustain exploration in T_{i+1}:

$$\mathbb{E}_{\pi_i}\left[E_{T_i}^{\text{terminal}}\right] < E_{\text{viable}}^{(i+1)}. \tag{1}$$

The agent then dies in the first few steps of the new task, observing only death trajectories and receiving no useful gradient signal. This differs fundamentally from catastrophic forgetting: forgetting is a *parameter-level* problem (old weights are overwritten), while viability failure is a *trajectory-level* problem (the agent never survives long enough to learn).

Table 1. Task catalog. All tasks share 4 movement actions and terminal-on-death. c_s: step cost. F: food count. H: hazard count.

Task	Grid	E_0	c_s	F	H	Key Challenge
Reach	5	15	1	1	0	Navigation
Recharge	5	8	1	1	0	Must eat to survive
Hazard Cross	5	12	1	1	2	Avoidance
Detour	5	9	1	1	2	Eat + avoid
Wall Maze	6	14	1	1	0	Spatial planning
Dual Food	6	10	1	2	1	Decision under scarcity
Haz. Gauntlet	6	14	1	1	4	Dense hazards
Conservation	5	20	2	1	1	High step cost
Endurance	7	16	1	1	2	Long path
Collect Exit	6	14	1	1	2	Gate mechanism
Sprint	5	14	1	1	0	Very short time limit
Gaunt. Refuel	7	16	1	2	4	All skills combined

EWC preserves old weights perfectly but cannot force the agent to conserve energy. ER replays old transitions from different energy regimes, providing misleading context. L2 regularization constrains parameter magnitude with no survival signal. All three operate at the wrong level of abstraction.

3.3 Homeostatic Drive Reduction

Following Keramati and Gutkin [1], we define an internal drive as distance from a homeostatic setpoint H^*: $d_t = |H^* - E_t|$. The HACE auxiliary reward is the one-step drive reduction:

$$r_t^H = |H^* - E_t| - |H^* - E_{t+1}|. \tag{2}$$

This signal is positive when actions bring the agent closer to homeostasis (e.g., collecting food when energy is low) and negative when actions increase deviation. It is *task-invariant* (unchanged across tasks), *dense* (per-step feedback), and *biologically motivated* (mirrors organismal drive reduction [1, 4]).

The full HACE reward combines this with the task-specific signal: $r_t^{\text{HACE}} = \alpha \cdot r_t^{\text{task}} + r_t^H$, where $\alpha = 1.0$ by default and $H^* = E_{\text{cap}}$. Because HACE operates at the *reward level* while EWC/ER operate at the *weight/data level*, the two are naturally orthogonal: Agent I (HACE+EWC) combines both to address viability and forgetting simultaneously.

4 Experimental Setup

4.1 Task Benchmark

We design 12 structurally diverse gridworld tasks sharing common energy dynamics but differing in spatial layout, hazards, food placement, walls, gates, step costs, and energy capacities. Table 1 summarizes the catalog. Each task is a Gymnasium environment: the agent moves on a discrete grid, consumes energy per step, collects food to restore energy, avoids hazard cells, and navigates walls and gate mechanisms. Episodes terminate on successful exit or energy depletion.

Table 2. Agent catalog. TR: task reward. EO: energy observation. HR: homeostatic reward. Seq: sequential (parameter carryover).

ID	Agent	TR	EO	HR	CRL	Seq
A	Task-only	✓	–	–	–	✓
B	Energy-aware	✓	✓	–	–	✓
C	**HACE**	✓	✓	✓	–	✓
D	Pure Homeostatic	–	✓	✓	–	✓
E	Task Oracle	✓	✓	–	–	–
F	EWC	✓	✓	–	EWC	✓
G	L2	✓	✓	–	L2	✓
H	Exp. Replay	✓	✓	–	ER	✓
I	**HACE+EWC**	✓	✓	✓	EWC	✓

We define three curated 10-task orderings: α **(Energy-Transfer)** where energy management skills build cumulatively, β **(Safety-Transfer)** where avoidance skills are learned first, and γ **(Interference)** with deliberately conflicting strategies (e.g., Conservation → Sprint). Full orderings are in Appendix B.

4.2 Agent Catalog

We compare 9 agent variants (Table 2) that systematically vary reward signal, energy observation, CRL method, and training mode. All share the same DQN architecture (2-layer MLP, 128 hidden units, ε-greedy; see Appendices A and D for details).

4.3 Metrics

We evaluate agents on: **current-task success rate** (fraction of evaluation episodes reaching exit), an **end-of-phase evaluation matrix** $R_{i,j}$ (success on task T_j after training through T_i), **boundary solvability** (whether the current policy can solve T_{i+1} from its actual terminal energy at T_i), **average terminal energy** at task boundaries, and standard **forgetting** $F_j = \max_{i \geq j} R_{i,j} - R_{n,j}$.

4.4 Protocol

All experiments use the α sequence under carryover boundary mode (the most challenging setting), with 600 episodes per task, 40 evaluation episodes every 50 training episodes, and 10 seeds per agent. We report means with standard deviations. Section 5.2 additionally compares reset vs. carryover.

5 Results

5.1 Full 9-Agent Suite (10-Task α, Carryover)

Figure 1 shows learning curves for all 9 agents across the 10-task sequence. To reduce visual clutter, we highlight the four primary agents—Task-Only (A), Hace (C), EWC (F), and Hace+EWC (I)—and render the remaining five in the background.

The learning curves reveal three key patterns. First, traditional CRL methods (EWC, L2, ER) provide no viability advantage over the unprotected Task-Only baseline in the later half of the sequence. On tasks 7–10 (Detour through Gauntlet Refuel), agents without homeostatic reward collapse to near-zero success. Second, Hace agents (C and I) consistently recover from initial dips at task boundaries, demonstrating that homeostatic reward provides a persistent survival signal that enables exploration in new tasks. Third, Hace+EWC (I) achieves the most consistent performance across the full sequence, combining the viability benefit of Hace with EWC's weight-protection on earlier tasks.

Figure 2 quantifies the viability mechanism directly. Panel (a) shows average boundary solvability: Hace and Hace+EWC achieve 3× higher boundary solvability than EWC, ER, or Task-Only. This confirms that viability failure is the bottleneck—CRL methods that only protect weights do not improve the agent's ability to survive the next task. Panel (b) reveals the underlying cause: Hace agents retain 3–5× more energy at task boundaries (11.2 and 9.1 vs. 2–4 for all other agents), directly enabling continued exploration.

Figure 3 presents the end-of-sequence success matrix. The heatmap makes the contrast stark: Task-Only (A) retains near-zero performance across all 10 tasks after reaching the end of the sequence, while Hace+EWC (I) maintains >0.25 success on Gauntlet Refuel—the hardest integrated task—and achieves the highest average success across all tasks (0.335). Even EWC (F) and ER (H), despite protecting earlier-task performance somewhat (Reach: 0.40–0.53), score zero on the final task. Hace alone (C) is the only non-oracle agent to achieve substantial success (>0.60) on Gauntlet Refuel, a task demanding all four skills (navigation, food collection, hazard avoidance, efficient energy use).

5.2 Reset vs. Carryover: Isolating Viability Failure

To confirm that the performance collapse under carryover is due to viability failure rather than forgetting, we ran a 3×2 factorial experiment (3 reward regimes × 2 boundary modes) on a 5-task pilot sequence, with 10 seeds per condition.

Figure 4 provides the direct evidence. Under reset mode, all three reward regimes achieve comparable performance: Task-Only reaches 0.45–0.75 success on Detour. Under carryover, Task-Only collapses to 0.30 while Hace maintains 0.80. Crucially, the *same network weights* produce these divergent outcomes—the only difference is whether energy resets between tasks. This rules out forgetting as the explanation and isolates viability failure as the true bottleneck. Homeostatic agents maintain higher terminal energy (2.95 vs. 0.35 for Task-Only at sequence end), which directly translates to survival in subsequent tasks.

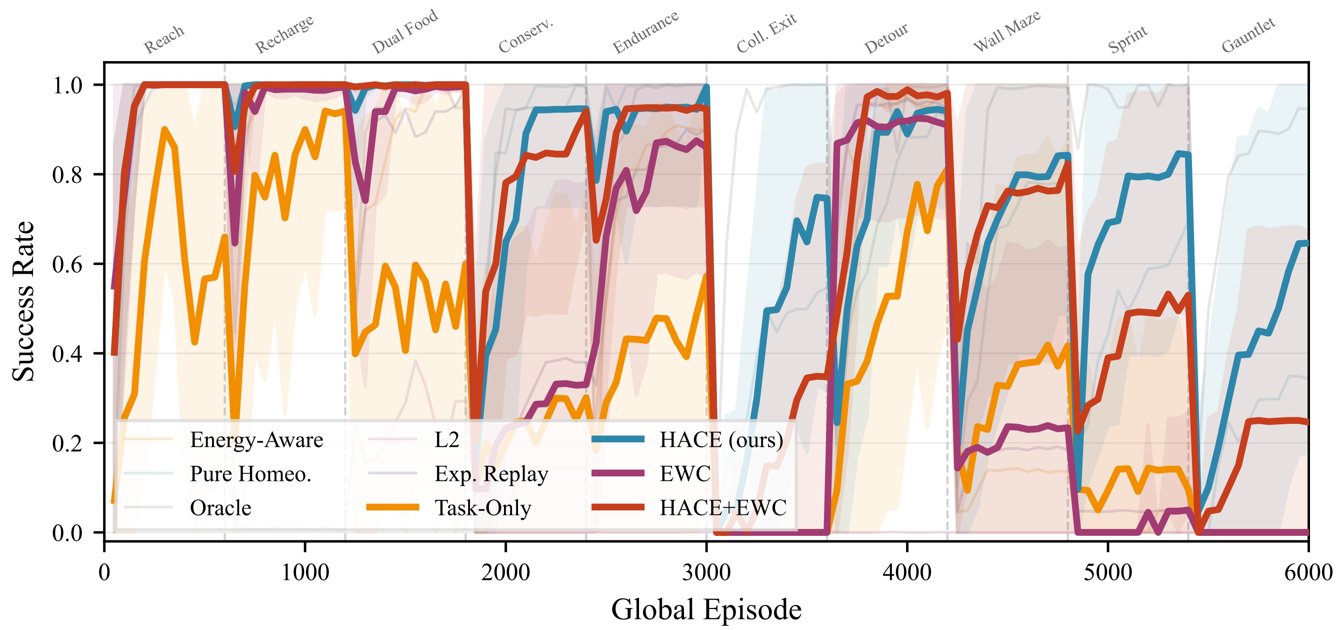

Figure 1. Current-task success rate across the 10-task α sequence (carryover). Task boundaries are marked by dashed lines. HACE (teal) and HACE+EWC (vermillion) consistently recover at task transitions, while Task-Only (amber) and EWC (mauve) collapse on later tasks. Background lines show remaining agents. Mean $\pm$ 1σ across 10 seeds.

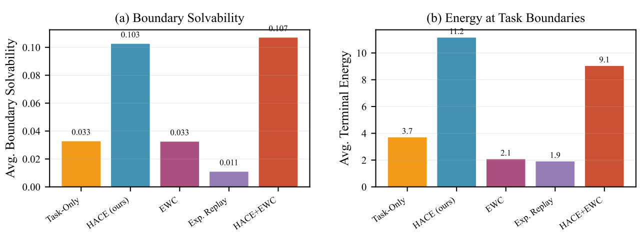

Figure 2. (a) Average boundary solvability across 9 task transitions. (b) Average terminal energy at task boundaries. HACE variants retain 3–5× more energy than all CRL baselines, directly enabling higher boundary solvability.

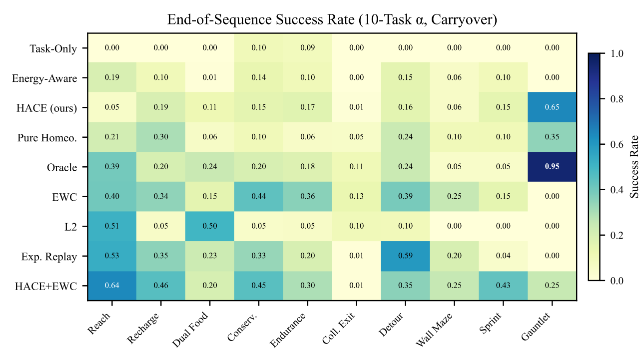

Figure 3. End-of-sequence success rate (agents × tasks) for the 10-task α sequence under carryover. Darker cells indicate higher success. HACE+EWC (bottom row) achieves the highest average performance; HACE (C) is the only agent to solve Gauntlet Refuel.

6 Discussion and Conclusion

Our results establish three findings that have implications for the broader CRL research agenda.

Viability failure is a distinct, overlooked failure mode. The reset-vs-carryover experiment (Section 5.2) unambiguously demonstrates that the performance collapse under carryover conditions is not catastrophic forgetting—the same weights succeed under reset. The full 9-agent suite confirms that established CRL methods (EWC, L2, ER) perform no better than unprotected baselines on viability metrics. This suggests that the standard CRL evaluation paradigm, which assumes infinite energy and no death, systematically overlooks the most acute failure mode in embodied settings.

HACE is simple, effective, and orthogonal to CRL methods. The HACE auxiliary reward (Equation 2) is a single-line modification to any RL training loop. It requires no task-specific tuning, no extra parameters, and no architectural changes—yet it produces 3× higher boundary solvability and enables agent C to be the only non-oracle agent solving Gauntlet Refuel (0.65 success). Agent I (HACE+EWC) achieves the best of both worlds: HACE-level viability with EWC's retention of earlier tasks, achieving the highest overall average success (0.335). This orthogonality suggests a

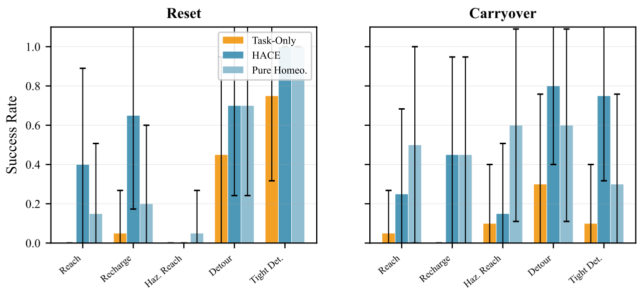

Figure 4. Final evaluation under reset (left) vs. carryover (right). Under reset, all agents perform comparably. Under carryover, Task-Only (amber) collapses while HACE (teal) maintains performance. The discrepancy proves the collapse is viability failure, not forgetting: the same weights work under reset.

principled decomposition of the embodied CRL problem: *staying alive* (reward level) and *remembering how* (weight level) are independent axes requiring independent solutions.

Limitations and future work. Our benchmark uses tabular gridworlds with vector observations; scaling to visual inputs and richer 3D environments is needed to confirm generality. All results use DQN, and replication with policy-gradient or Transformer-based architectures would further validate the finding. We model a single resource (energy), whereas real agents manage multiple coupled drives. Finally, we fix $H^* = E_{\text{cap}}$; learning or adapting the setpoint per task remains unexplored. Despite these limitations, the conceptual contribution—that viability failure is a distinct bottleneck that CRL methods cannot address—holds independently of environment complexity, and HACE's simplicity makes it immediately applicable to any embodied sequential RL setting with shared resource constraints.

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A Hyperparameters

Table 3 lists all hyperparameters shared across agents. Agent-specific parameters (EWC λ, ER reservoir size) are also included.

Table 3. Training hyperparameters.

Parameter	Value
Architecture	2-layer MLP, 128 hidden units
Optimizer	Adam
Learning rate	5×10^{-4}
Discount γ	0.99
Batch size	64
Replay buffer	20,000 transitions
Target network update	every 25 episodes
ε-greedy	1.0 → 0.05 over 300 episodes
Episodes per task	600
Evaluation	40 episodes every 50 training episodes
EWC λ (F, I)	400.0
EWC Fisher samples	1,000
L2 λ (G)	100.0
ER reservoir (H)	5,000 transitions, ratio 0.5
HACE setpoint	E_{cap} per task
HACE coefficient	1.0
Task reward coefficient α	1.0

B Task Sequence Definitions

Table 4 lists the three curated 10-task orderings used in our experiments. Sequence α is the primary ordering used in all main-text experiments; β and γ are reserved for future cross-sequence generalization analyses.

Table 4. Full 10-task sequence orderings.

Seq.	Task Order
α	Reach → Recharge → Dual Food → Conservation → Endurance → Collect Exit → Detour → Wall Maze → Sprint → Gauntlet Refuel
β	Reach → Hazard Cross → Haz. Gauntlet → Wall Maze → Detour → Recharge → Sprint → Conservation → Collect Exit → Gauntlet Refuel
γ	Conservation → Sprint → Haz. Gauntlet → Reach → Endurance → Dual Food → Hazard Cross → Collect Exit → Recharge → Wall Maze

C Preliminary Experiment: 5-Task Sequential Pilot

Before deploying the full 9-agent suite, we conducted a smaller-scale pilot to validate the core hypothesis. We trained two agents—a task-specific agent (comparable to Agent A/B) and a homeostatic agent (comparable to Agent D)—on a 5-task legacy sequence: Reach → Recharge → Hazard Cross → Detour → Sprint, with 600 episodes per task and 10 seeds.

The pilot (Figure 5) revealed a consistent tradeoff: task-specific reward learns early, simple tasks faster (Reach: 0.6 within 200 episodes vs. 400 for homeostatic), but the homeostatic agent matches or exceeds performance on later tasks requiring resource management and hazard avoidance. Final retention on Hazard Cross was 0.90 vs. 0.70, and on Detour 0.80 vs. 0.60. This preliminary finding motivated the full-scale 9-agent factorial design presented in the main text.

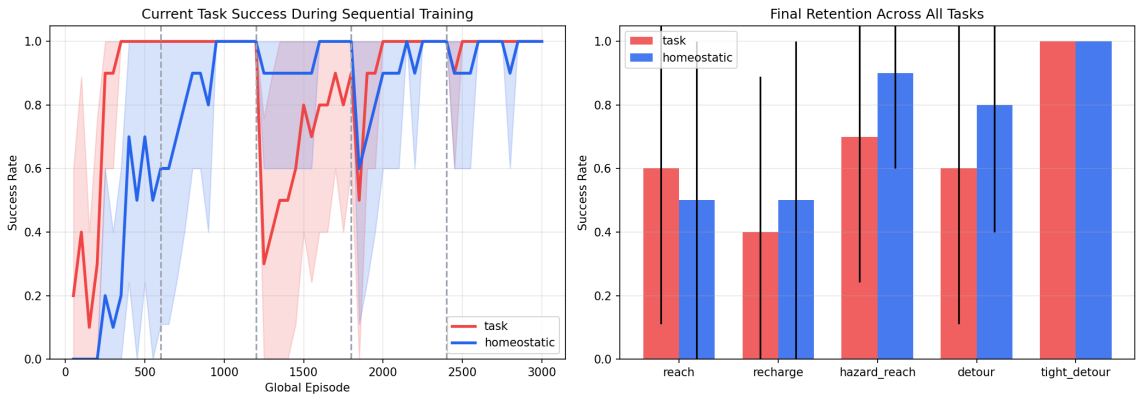

Figure 5. 5-task sequential pilot. *Left*: Current-task success rate during sequential training. The task-specific agent (red) learns early tasks faster; the homeostatic agent (blue) matches or exceeds on later tasks. *Right*: Final retention across all five tasks. The homeostatic agent shows stronger retention on resource-dependent tasks (Hazard Cross: 0.90 vs. 0.70; Detour: 0.80 vs. 0.60).

D Observation Space Details

All agents receive the same 19-dimensional vector observation (Table 5). The observation encodes the agent's position, goal location, food and hazard positions, and—critically—the agent's current energy level. All spatial features are normalized by grid size; energy is normalized by the task's energy cap.

Table 5. Observation vector layout. All spatial features are normalized to $[0, 1]$. Agent A receives a masked energy value (-1) to isolate the effect of reward signal from observation content.

Indices	Content	Normalization
0–1	Agent position (row, col)	/ (grid_size − 1)
2–3	Exit position (row, col)	/ (grid_size − 1)
4–6	Food 1: row, col, available	/ grid; boolean
7–9	Food 2: row, col, available	/ grid; boolean
10	Energy level	/ energy_cap (or −1 if masked)
11	Task index	/ (num_tasks − 1)
12–17	Hazard positions (up to 3)	/ grid (−0.25 if absent)
18	Gate locked	Boolean